

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int SOUND_PIN = A0;
const int BUTTON_PIN = 2;

const int THRESHOLD = 7;

void setup() {
    Serial.begin(9600);

    lcd.init();
    lcd.backlight();

    pinMode(BUTTON_PIN, INPUT_PULLUP);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("PAPPADAM TEST");
    lcd.setCursor(0, 1);
    lcd.print("PRESS START");
}

void loop() {

    // Wait for START button
    if (digitalRead(BUTTON_PIN) == LOW) {

        delay(50);    // button debounce

        // Start screen
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("TEST STARTED");
        lcd.setCursor(0, 1);
        lcd.print("MAKE A CRUNCH!");

        delay(1000);

        // Get background sound level
        int baseline = analogRead(SOUND_PIN);
```

```
int maxChange = 0;

// Listen for 3 seconds
unsigned long startTime = millis();

while (millis() - startTime < 3000) {

    int soundValue = analogRead(SOUND_PIN);

    int change = abs(soundValue - baseline);

    if (change > maxChange) {
        maxChange = change;
    }

    Serial.print("Sound Change: ");
    Serial.println(change);

    delay(10);
}

// Show result
lcd.clear();

if (maxChange >= THRESHOLD) {

    lcd.setCursor(0, 0);
    lcd.print("CRUNCH DETECTED");

    lcd.setCursor(0, 1);
    lcd.print("SCORE: ");
    lcd.print(maxChange);

} else {

    lcd.setCursor(0, 0);
    lcd.print("NO CRUNCH");

    lcd.setCursor(0, 1);
    lcd.print("SCORE: ");
    lcd.print(maxChange);

}
```

```
    delay(3000);

    // Ready for next test
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("PAPPADAM TEST");
    lcd.setCursor(0, 1);
    lcd.print("PRESS START");

    // Wait for button release
    while (digitalRead(BUTTON_PIN) == LOW) {
        delay(10);
    }
}
```